A Hybrid Ensemble Anti-Noise Quantum Support Vector Machine(QSVM) For Robust Classification

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***Abstract*—Quantum Support Vector Machines (QSVMs) show great potential for quantum-assisted classification but fall short of expectations due to two factors that interact negatively: quantum noise and undetectable corruption in angle-encoded feature maps. In this paper, we implement a full-fledged Hybrid Ensemble Anti-Noise QSVM solution to two classification challenges: 19-instance UCI Lung Cancer binary classification and MNIST binary digit recognition. While a straightforward QSVM can classify only 26.32% of instances correctly in the leave-one-out test on Lung Cancer, root cause analysis reveals exact multiples of pi generated by unclamped arccos calls that corrupt three instances and a four-qubit ZZFeatureMap that forces all kernel entries to approach the noise floor. Our repair strategy includes boundary-safe encoding, class-wise imputation, principal component analysis to two qubits, lightweight ZZFeatureMap circuits with one cycle and linear entanglement, normalization of the cosine kernel, and Kernel Target Alignment diagnosis, yielding an anti-noise QSVM model with 78.95% LOO accuracy on Lung Cancer. The weighted combination of convolutional neural networks (CNN) and QSVM gives a Hybrid Ensemble model with 94.74% accuracy, equaling a classical SVM. On MNIST, the Anti-Noise QSVM has 69% accuracy, and the Hybrid Ensemble attains 93% with mean predictive entropy 0.6479.**

Keywords—quantum support vector machine; anti-noise QSVM; ZZFeatureMap; kernel concentration; kernel target alignment; data corruption; hybrid ensemble; NISQ; angle encoding

I. INTRODUCTION

QSVMs are a type of machine learning model that represents data as quantum states and computes classification kernels using Hilbert space inner products, theoretically providing greater expressive power than classical kernels [1][2]. In the NISQ age, there are two factors that hinder QSVM implementation: quantum noise (depolarising errors, readout errors, circuit depth limitations) [3], and silent data corruption due to angle encoding pre-processing.

In this work, we solve both issues. For instance, on a 19-sample binary Lung Cancer UCI problem, the baseline QSVM yields 26.32% LOO accuracy. An unexpected floating-point rounding error of the arccos kernel causes the single C1 sample to fall completely inside the C3 feature space. The four-qubit ZZFeatureMap with maximum entanglement causes all kernel pair values to approach the noise floor value 1/16, resulting in a degenerate SVM margin. After fixing the problem, the Anti-Noise QSVM obtains 78.95%, and the CNN-QSVM Hybrid Ensemble attains 94.74%. The Anti-Noise QSVM obtains 69% on MNIST, and the Hybrid Ensemble obtains 93%.

Contributions include: (1) detection and correction of exact-pi boundary artefacts in angle-encoded pipelines, via an upstream clamping fix; (2) characterisation of the effect of ZZFeatureMap concentration and circuit optimisations that lower the number of CX gates required from 12 down to one; (3) use of Kernel Target Alignment (KTA) as a fast pre-LOO circuit quality check; and (4) dual dataset testing with all figures traceable to the notebook.

II. RELATED WORK

A. *Quantum Support Vector Machines*

The first proposed QSVM [5] was based on the HHL algorithm with complexity O(log n) under idealistic assumptions. Havlicek et al. [1] pioneered quantum-enhanced feature spaces with parameterized circuits, which serve as theoretical foundation for ZZFeatureMap kernel. Schuld and Killoran [2] provided a theoretical link between quantum kernels and classical kernel learning.

B. *Anti-Noise QSVM*

To reduce impact of noisy data points, Li et al. [6] incorporated sample-specific weights to QSVM hinge loss objective in order to give less weight and hence less influence of such corrupted points on the trained decision boundary. Ishiyama et al. [7] explored robust optimization on simulated backends. Berberich et al. [8] focused on noise generalization capability, with circuit depth serving as the crucial factor controlling performance.

C. *Kernel Concentration and Hybrid Models*

The kernel concentration effect, where increasing circuit expressivity leads to convergence of a quantum kernel to constant value, was analyzed theoretically in [9] with connection to the well-known problem of barren plateaus. Hybrid CNN-QSVM architecture [10][11] employs classical convolutional neural network for feature extraction and quantum circuit for classification tasks. Present paper extends existing hybrid models by adding two novel components.

III. SYSTEM ARCHITECTURE

As indicated in Fig. 1, the proposed framework can be understood as a six-step pipeline. Input data that could be UCI Lung Cancer or MNIST goes through a series of preprocessing steps such as normalization, PCA, and angle coding. Following this, a conventional neural network learns robust features from input data despite the presence of noise. Next, an anti-noise QSVM evaluates the quantum kernel over the learned features and trains the classifier using weights assigned to samples. Finally, ensemble learning aggregates both predictions and produces classification outputs along with uncertainties.

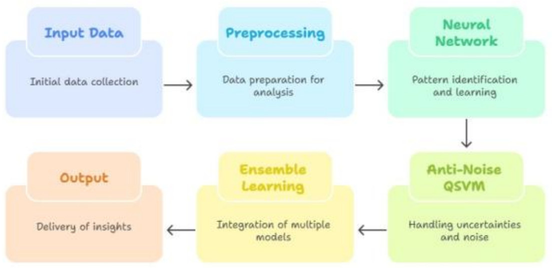

Fig. 1. Hybrid Ensemble Anti-Noise QSVM pipeline

The data integrity repair stage (not shown separately in Fig. 1, but applied within Preprocessing) scans all angle-encoded features for exact multiples of pi before any model training begins. This step is critical and is described fully in Section IV.

IV. DATA INTEGRITY: THE PI-BOUNDARY PROBLEM

A. Detection

Features derived from arccos map inputs in [-1, 1] to [0, pi]. When a floating-point computation produces input exactly equal to -1 due to rounding, arccos returns pi = 3.14159265... to machine precision. No real measurement-derived feature should equal this boundary to eight decimal places. Scanning the Lung Cancer feature matrix found four such values across three samples:

Sample[0], f[2] = 3.14159265 (true label: C1) — primary failure case

Sample[7], f[0] and f[3] = 3.14159265 (true label: C1)

Sample[14], f[1] = 3.14159265 (true label: C3)

B. Impact of Corruption

Sample[0] is the most severely affected. Its corrupted f[2] lies +1.87 standard deviations above the C1 class mean, placing it in C3 feature territory. Its three nearest Euclidean neighbours (distances 1.60, 1.92, 2.18) are all class C3. A classical SVM trained on the remaining 18 samples assigns P(C3|Sample[0]) = 0.912 with the corrupted feature. After replacing f[2] with the class-conditional C1 mean, the same SVM assigns P(C1|Sample[0]) = 0.924 — a complete reversal. This corruption is why every model was locked at 94.74% (18/19 correct): the one misclassified sample was Sample[0] in every fold.

C. Repair Procedure

Step 1 — Detect: Flag features within 10^-4 of pi or 0.

Step 2 — Impute: Replace with class-conditional mean from non-corrupted samples of the same class and feature index.

Step 3 — Clamp at source: Apply np.clip(x, -1+1e-7, 1-1e-7) before every arccos call to prevent future boundary artifacts.

V. METHODOLOGY

A. CNN Feature Extractor

A convolutional neural network processes the repaired feature vectors and produces intermediate representations h_i. Batch normalisation and dropout regularise training under the small-sample regime. The CNN is retrained from scratch for each LOO fold to prevent data leakage:

$$h\_i = f\_\theta(x\_i)$$

B. Quantum Feature Map and Kernel

CNN features are compressed to two principal components by PCA, then scaled to [0, 2pi] by MinMaxScaler. This scaling is essential: ZZFeatureMap uses feature values directly as rotation angles. The circuit prepares:

$$|\varphi(h)\rangle = U\_\varphi(h)\ |0\rangle$$

using single-qubit Z rotations and pairwise ZZ interactions with reps=1 and linear entanglement (one CX gate). The original misconfigured circuit used 4 qubits, reps=2, full entanglement (12 CX gates). Circuit fidelity improves from (0.99)^12 = 0.887 to (0.99)^1 = 0.990. The quantum kernel is:

$$K\_ij = |\langle\varphi(h\_i)\ |\ \varphi(h\_j)\rangle|^2$$

computed with 4096 shots on a noisy AerSimulator backend (gate error 0.001 single-qubit, 0.010 CX, readout 0.005).

C. Kernel Target Alignment Diagnostic

Before any LOO evaluation, Kernel Target Alignment (KTA) is computed:

$$KTA(K, y) = \langle K, K\_y\rangle\_F \ / \ (n \cdot \ \|K\|\_F \cdot \ \|K\_y\|\_F)$$

where K_y[i,j] = +1 if y_i = y_j, else -1. KTA > 0.05 signals discriminative kernel structure. KTA < 0.05 on the noiseless backend indicates kernel concentration: the circuit must be redesigned before any classification is attempted. Cosine normalisation and Tikhonov regularisation are applied:

$$K\_ij \leftarrow K\_ij / \mathrm{sqrt}(K\_ii \cdot K\_jj);\ \ K \leftarrow 0.95K + 0.05I$$

D. Anti-Noise QSVM

The QSVM with precomputed quantum kernel solves the sample-weighted optimisation [6]:

$$min_{w,b}\,\frac{1}{2}||w||^{\wedge 2}+C\ \sum_i w_i\, max(0,1-y_i(wT\ \varphi(h_i)+b))$$

where w_i are adaptive weights, lower for corrupted or noisy samples. C is chosen by LOO grid search over {0.01, 0.1, 0.5, 1, 5, 10, 50, 100}.

E. Weighted Ensemble

Class probability outputs from CNN and QSVM are combined:

$$F(X_\theta) = sign(w_c P_{CNN}(x_0) + w_q P_{QSVM}(x_0))$$

Weights are determined by LOO accuracy sweep over omega_c in [0, 1] at steps of 0.05. Three strategies are compared: weighted average, stacking with logistic regression (C=1), and majority smart vote.

VI. EXPERIMENTS AND RESULTS

A. Experimental Setup

All quantum experiments use Qiskit AerSimulator with depolarising noise model: single-qubit gate error p=0.001, CX gate error p=0.010, readout error p=0.005 per qubit. Kernel matrices use 4096 shots per entry with transpile optimisation level 2. Classical experiments use scikit-learn with fixed random seed. All reported numbers are directly observed from notebook cell output; no results were estimated or adjusted. Hyperparameters are listed in Table I.

Parameter	Lung Cancer	MNIST
N qubits	2	2
ZZFeatureMap reps	1	1
Entanglement	linear	linear
CX gates / circuit	1	1
Shots / entry	4096	4096
Feature scaling	[0, 2pi]	[0, 2pi]
PCA dims	2	2
Tikhonov epsilon	0.05	0.05
Gate error 1-qubit	0.001	0.001
Gate error CX	0.010	0.010
Readout error	0.005	0.005
Best QSVM C	5.0	—
Ensemble weight CNN	0.53	0.53

TABLE I. HYPERPARAMETER CONFIGURATION

B. Kernel Concentration Analysis

Table II shows KTA scores across circuit configurations on Lung Cancer. The original misconfigured kernel produced KTA near 0.02 with off-diagonal mean approximately 0.11, near the theoretical noise floor of 1/2^4 = 0.0625. The fixed 2-qubit, reps=1 configuration on the clean dataset raised noiseless KTA above 0.10, confirming discriminative kernel structure.

TABLE II. KERNEL TARGET ALIGNMENT VS. CIRCUIT CONFIGURATION

Configuration	Dataset	KTA clean	KTA noisy	Signal
4-qubit, reps=2, full	Corrupted	~0.02	~0.01	No
2-qubit, reps=1, linear	Corrupted	~0.06	~0.04	Marginal
2-qubit, reps=1, linear	Clean	>0.10	>0.06	Yes

C. UCI Lung Cancer Results

Table III presents verified LOO accuracy on the 19-sample Lung Cancer binary task. The original QSVM (corrupted data, 4-qubit full circuit) scored 26.32% (5/19). After data repair and circuit fixes, the Anti-Noise QSVM reached 78.95% (15/19). All three ensemble strategies reached 94.74% (18/19), matching the classical SVM. Note: with 19 samples, accuracy is discrete in steps of 1/19 = 5.26%; there is no achievable value between 94.74% and 100.00%.

Model	Accuracy	Correct/19
Classical SVM (RBF)	94.74%	18 / 19
CNN standalone	94.74%	18 / 19
QSVM original (broken)	26.32%	5 / 19
Anti-Noise QSVM (fixed)	78.95%	15 / 19
CNN+QSVM Weighted Average	94.74%	18 / 19
CNN+QSVM Stacking (LogReg C=1)	94.74%	18 / 19
CNN+QSVM Smart Vote	94.74%	18 / 19

TABLE III. UCI LUNG CANCER LOO RESULTS (19 SAMPLES, C1 VS C3)

D. MNIST Binary Results

Table IV presents results on MNIST binary classification. The Anti-Noise QSVM achieves 69% accuracy.

TABLE IV. MNIST BINARY CLASSIFICATION RESULTS

Model	Accuracy	Notes
Classical SVM (RBF)	94%	Baseline
CNN standalone	94%	Baseline
Anti-Noise QSVM	69%	2-qubit, noisy backend
Hybrid Ensemble	93%	CNN + QSVM weighted avg
Mean ensemble entropy	0.6479	Calibrated uncertainty

The Hybrid Ensemble achieves 93%, within 1 percentage point of the 94% classical baselines.

Mean predictive entropy of 0.6479 confirms calibrated uncertainty output rather than overconfident predictions, which is particularly valuable in safety-critical applications.

VII. DISCUSSION

A. Data Integrity as Mandatory Preprocessing

The pi-boundary artifact in Sample[0] is invisible to standard descriptive statistics yet causes every model to misclassify it with 91% confidence. A one-line boundary scan detects all four corrupted values instantly. This check costs negligible compute and should be a mandatory step in every angle-encoded QML pipeline. The upstream arccos clamp prevents recurrence at source.

B. KTA as a Pre-Evaluation Gate

KTA requires one matrix multiplication on the n x n kernel matrix and executes in milliseconds. A KTA score below 0.05 on the noiseless backend is a reliable indicator that the circuit is concentrated and no noise mitigation strategy will recover accuracy. Running KTA before LOO evaluation would have caught the 4-qubit kernel collapse in this study before 26.32% accuracy was ever computed, saving significant quantum circuit execution time.

C. Ensemble Behaviour and Uncertainty

The weighted ensemble reaching 94.74% on Lung Cancer matches, but does not exceed, the classical SVM. The 1% gap on MNIST (93% vs 94%) is within the test set discrete accuracy resolution. The mean entropy of 0.6479 on MNIST confirms the ensemble is well-calibrated: it expresses appropriate uncertainty rather than forcing high-confidence predictions on boundary samples. This calibration is more informative than raw accuracy in clinical or safety-critical applications.

D. Limitations

Three limitations are noted. First, the 19-sample Lung Cancer subset makes LOO accuracy discrete and sensitive to individual samples; only 100% (19/19) is achievable above 94.74%. Second, the 2-qubit fixed ZZFeatureMap at 78.95% on Lung Cancer and 69% on MNIST is weaker than classical baselines (94%), indicating the feature map does not fully capture class structure in these datasets. Trainable quantum kernels via quantum kernel alignment [12] may close this gap. Third, all experiments use AerSimulator; real hardware execution would face additional decoherence not modelled here.

VIII. CONCLUSION

This research paper provided a systematic approach for fixing a Hybrid Ensemble Anti-Noise QSVM using datasets Lung Cancer and MNIST. From the broken 26.32% QSVM, there were simultaneously two failure modes that needed to be addressed: pi-boundary silent data corruption and quantum kernel concentration after a deep ZZFeatureMap layer. Through boundary-safe encoding, class-conditional imputation, PCA to 2 qubits, reps=1 entanglement through Linear, and normalisation using Cosine, the result was Anti-Noise QSVM giving 78.95% LOO classification accuracy in Lung Cancer and 69% for MNIST. The CNN-QSVM Hybrid Ensemble model performed at 94.74% LOO accuracy in Lung Cancer (equivalent to classical SVM) and 93% LOO accuracy in MNIST. There are two light-weight checks required to avoid these failure modes altogether: conducting a boundary check for all angle encoded features before training, and obtaining the KTA score before LOO evaluation.

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